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Aragya Goyal

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Education

- **University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
  B.S. - Computer Engineering (Autonomous Systems Focus); **GPA: 3.99/4.00** *August 2022 - April 2026*
  Courses: Robotic Control, Data Structures and Algorithms, Embedded Processors, Microelectronics

Professional Experience

- **ESTAT Actuation** **Pittsburgh, PA**
  Robotics Developer Co-op *May 2025 - Present*
  - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
  - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
  - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.
- **Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
  Undergraduate Roboticist Researcher *April 2023 - Present*
  - **ZOË 2 Rover**: (tinyurl.com/zoe2rover)
    - Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
    - Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
  - **Underwater Snake Robot**: (tinyurl.com/humrsCMU)
    - Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
    - Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
    - Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
  - **Apple's E-Waste Recycling Project**: (tinyurl.com/applecmu)
    - Labeled datasets consisting of 4000+ images for Machine Learning Models to detect screws in e-waste images.
    - Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
    - Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

Leadership & Activities

- **Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
  Chief Engineer/Integration Lead *August 2022 - Present*
  - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams. (tinyurl.com/roverimages)
  - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
  - Securing and managing over $7,000 in funding to support team development and future growth.
  - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)

Projects

- **Micromouse Robot**: Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. (tinyurl.com/goyalmm)
- **555 Timer Roulette Board**: Designed a PCB LED Roulette Wheel using 555 timer and CD4017 counter, featuring dynamic speed control using capacitive timing and a 3D-printed enclosure. (tinyurl.com/555roulette)
- **VoltVitals Embedded Board**: Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. (tinyurl.com/voltvitals)

Publications

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal,** A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, **"BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,"** in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)

Skills and Awards

- **Software**: Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM and RISC-V Assembly, CANOpen, Google Colab
- **Electrical**: Altium, KiCAD, LTSpice, Soldering, Arduino, ESP32
- **Mechanical**: Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
- **Awards**: Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, Eagle Scout